

# Domain-Specific NLP

# Introduction

**Pretrained (Large) Language Models** are trained on content crawled over the internet, books, reports and news papers and are, hence **are open-domain**.

A **textual domain** is the **distribution over language characterizing a given topic or genre** [1].

- You are more likely to see the word "integer" in computer science than in news papers.

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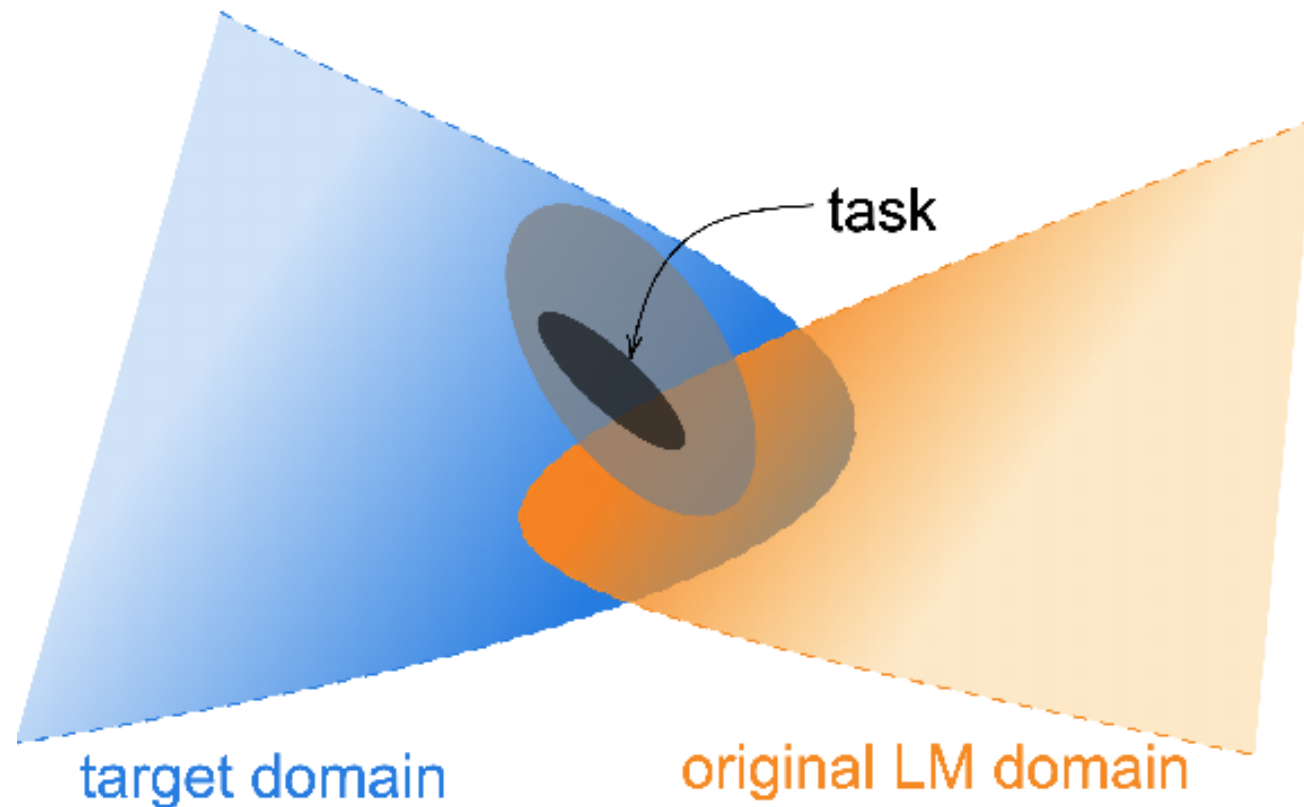
## 3. **Learning Long-Range Dependencies**

- a. Long-range attention models
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# Domain-Specific Models

# ***Don't stop pre-training***

*Don't Stop Pretraining: Adapt Language Models to Domains and Tasks. [1]*



## Don't stop pre-training

| Domain             | Pretraining Corpus                                   | # Tokens | Size  | $\mathcal{L}_{\text{ROB.}}$ | $\mathcal{L}_{\text{DAPT}}$ |
|--------------------|------------------------------------------------------|----------|-------|-----------------------------|-----------------------------|
| BIOMED             | 2.68M full-text papers from S2ORC (Lo et al., 2020)  | 7.55B    | 47GB  | 1.32                        | 0.99                        |
| CS                 | 2.22M full-text papers from S2ORC (Lo et al., 2020)  | 8.10B    | 48GB  | 1.63                        | 1.34                        |
| NEWS               | 11.90M articles from REALNEWS (Zellers et al., 2019) | 6.66B    | 39GB  | 1.08                        | 1.16                        |
| REVIEWS            | 24.75M AMAZON reviews (He and McAuley, 2016)         | 2.11B    | 11GB  | 2.10                        | 1.93                        |
| ROBERTA (baseline) | see Appendix §A.1                                    | N/A      | 160GB | ‡1.19                       | -                           |

**Table 1:** List of the domain-specific unlabeled datasets. In columns 5 and 6, we report ROBERTA’s masked LM loss on 50K randomly sampled held-out documents from each domain before ( $\mathcal{L}_{\text{ROB.}}$ ) and after ( $\mathcal{L}_{\text{DAPT}}$ ) *DAPT* (lower implies a better fit on the sample). ‡ indicates that the masked LM loss is estimated on data sampled from sources similar to ROBERTA’s pretraining corpus.

## ***Don't stop pre-training***

|         |       |       |         |        |       |
|---------|-------|-------|---------|--------|-------|
| PT      | 100.0 | 54.1  | 34.5    | 27.3   | 19.2  |
| News    | 54.1  | 100.0 | 40.0    | 24.9   | 17.3  |
| Reviews | 34.5  | 40.0  | 100.0   | 18.3   | 12.7  |
| BioMed  | 27.3  | 24.9  | 18.3    | 100.0  | 21.4  |
| CS      | 19.2  | 17.3  | 12.7    | 21.4   | 100.0 |
|         | PT    | News  | Reviews | BioMed | CS    |

**Figure 2:** Vocabulary overlap (%) between domains. PT denotes a sample from sources similar to ROBERTA's pretraining corpus. Vocabularies for each domain are created by considering the top 10K most frequent words (excluding stopwords) in documents sampled from each domain.

## Don't stop pre-training

| Domain  | Task          | RoBERTa             | Additional Pretraining Phases |                            |                            |
|---------|---------------|---------------------|-------------------------------|----------------------------|----------------------------|
|         |               |                     | DAPT                          | TAPT                       | DAPT + TAPT                |
| BioMed  | CHEMPROT      | 81.9 <sub>1.0</sub> | 84.2 <sub>0.2</sub>           | 82.6 <sub>0.4</sub>        | <b>84.4</b> <sub>0.4</sub> |
|         | †RCT          | 87.2 <sub>0.1</sub> | 87.6 <sub>0.1</sub>           | 87.7 <sub>0.1</sub>        | <b>87.8</b> <sub>0.1</sub> |
| CS      | ACL-ARC       | 63.0 <sub>5.8</sub> | 75.4 <sub>2.5</sub>           | 67.4 <sub>1.8</sub>        | <b>75.6</b> <sub>3.8</sub> |
|         | SciERC        | 77.3 <sub>1.9</sub> | 80.8 <sub>1.5</sub>           | 79.3 <sub>1.5</sub>        | <b>81.3</b> <sub>1.8</sub> |
| NEWS    | HYPERPARTISAN | 86.6 <sub>0.9</sub> | 88.2 <sub>5.9</sub>           | <b>90.4</b> <sub>5.2</sub> | 90.0 <sub>6.6</sub>        |
|         | †AGNEWS       | 93.9 <sub>0.2</sub> | 93.9 <sub>0.2</sub>           | 94.5 <sub>0.1</sub>        | <b>94.6</b> <sub>0.1</sub> |
| REVIEWS | †HELPFULNESS  | 65.1 <sub>3.4</sub> | 66.5 <sub>1.4</sub>           | 68.5 <sub>1.9</sub>        | <b>68.7</b> <sub>1.8</sub> |
|         | †IMDB         | 95.0 <sub>0.2</sub> | 95.4 <sub>0.1</sub>           | 95.5 <sub>0.1</sub>        | <b>95.6</b> <sub>0.1</sub> |

**Table 5:** Results on different phases of adaptive pretraining compared to the baseline RoBERTa (col. 1). Our approaches are *DAPT* (col. 2, §3), *TAPT* (col. 3, §4), and a combination of both (col. 4).



## Specialized models (BioBERT, SciBERT, Galactica)

"[..] the word distributions of general and biomedical corpora are quite different, which can often be a problem for biomedical text mining models." [2]

# Specialized models (BioBERT, SciBERT, Galactica)

: 1. List of text corpora used for BioBERT

| Corpus                 | # of words (B) | Domain     |
|------------------------|----------------|------------|
| English Wikipedia      | 2.5B           | General    |
| BooksCorpus            | 0.8B           | General    |
| PubMed Abstracts       | 4.5B           | Biomedical |
| PMC Full-text articles | 13.5B          | Biomedical |

## Specialized models (BioBERT, SciBERT, Galactica)

"We showed that **pre-training BERT on biomedical corpora is crucial in applying it to the biomedical domain**. Requiring minimal task-specific architectural modification, **BioBERT outperforms previous models on biomedical text mining tasks** such as NER, RE and QA."

# Specialized models (BioBERT, SciBERT, Galactica)

| Task | Dataset      | BIOBERT | SCI-BERT |
|------|--------------|---------|----------|
| NER  | BC5CDR       | 88.85   | 90.01    |
|      | JNLPBA       | 77.59   | 77.28    |
|      | NCBI-disease | 89.36   | 88.57    |
| REL  | ChemProt     | 76.68   | 83.64    |

**Table 2:** Comparing SciBERT with the reported BioBERT results on biomedical datasets.

# Specialized models (BioBERT, SciBERT, Galactica)

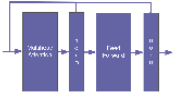
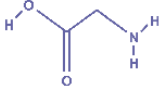
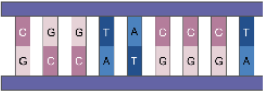
**NB:** SciBERT was trained on curated textual data ; not trained on code or script for example---at least not trained directly and purposefully on this kind of data

# Specialized models (BioBERT, SciBERT, Galactica)

"Unlike search engines, language models can potentially store, combine and reason about scientific knowledge." [4]

- Specialized models (BioBERT, SciBERT, Galactica) were trained on a rather small highly curated dataset.
- The data was standardized in markdown format.

# Specialized models (BioBERT, SciBERT, Galactica)

| Modality     | Entity                 | Sequence                                  |                                                                                     |
|--------------|------------------------|-------------------------------------------|-------------------------------------------------------------------------------------|
| Text         | Abell 370              | Abell 370 is a cluster...                 |  |
| LaTeX        | Schwarzschild radius   | $r_{\{s\}} = \frac{2GM}{c^2}$             | $r_s = \frac{2GM}{c^2}$                                                             |
| Code         | Transformer            | <code>class Transformer(nn.Module)</code> |  |
| SMILES       | Glycine                | <chem>C(C(=O)O)N</chem>                   |  |
| AA Sequence  | Collagen α-1(II) chain | MIRLGAPQTL..                              |  |
| DNA Sequence | Human genome           | CGGTACCCTC..                              |  |

**Table 1: Tokenizing Nature.** Galactica trains on text sequences that represent scientific phenomena.

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# Specialized models (BioBERT, SciBERT, Galactica)

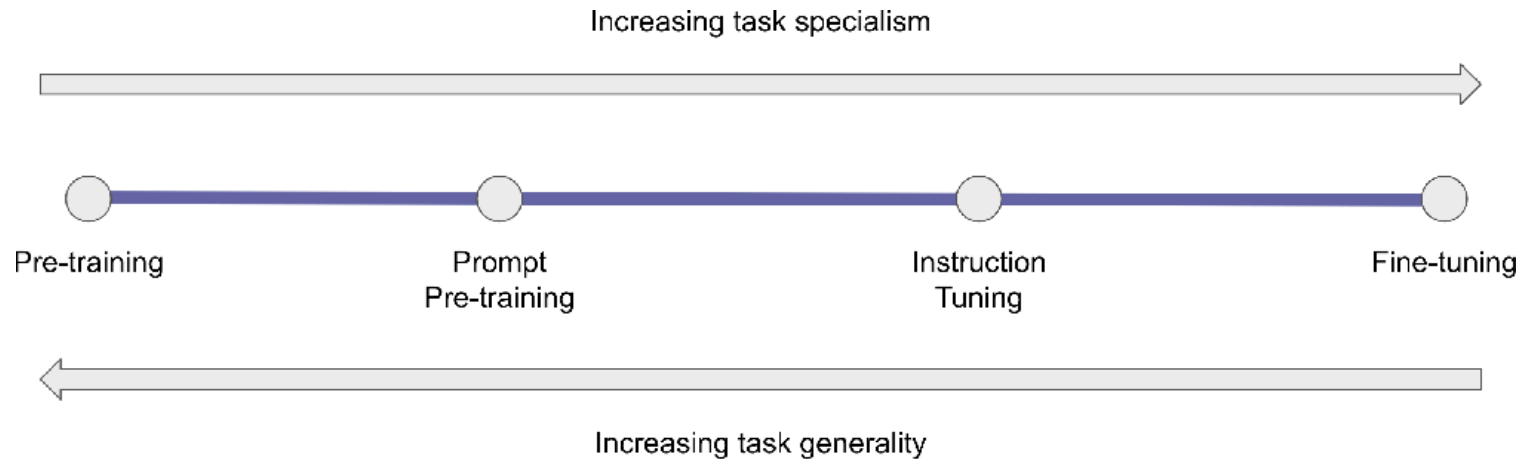
1. **Citations:** wrapped with special reference tokens [START\_REF] and [END\_REF].
2. **Step-by-Step Reasoning:** wrapped with a working memory token `<work>`, mimicking an internal working memory context.
3. **Mathematics:** for mathematical content, with or without LaTeX, ASCII operations are splitted into individual characters. Parentheses are treated like digits. The rest of the operations allow for unsplit repetitions. Operation characters are `!"#$%&'*+,-./:;<=>?^_`|` and parentheses are `()[]{}.`



4. **Numbers:** splitted into individual tokens. For example 737612.62 -> 7,3,7,6,1,2,,6,2.
5. **SMILES formula:** wrapped with [START\_SMILES] and [END\_SMILES] and tokenized absed on characters. Similarly [START\_I\_SMILES] and [END\_I\_SMILES] is used where isomeric SMILES is denoted.
6. **Amino acid sequences:** wrapped with [START\_AMINO] and [END\_AMINO] and apply character-based tokenization, treating each amino acid character as a single token. For example, MIRLGAPQTL -> M,I,R,L,G,A,P,Q,T,L.

1. **DNA sequences:** tokenized based on characters and wrapped inside [START\_DNA] and [END\_DNA]. For example, CGGTACCCCTC -> C, G, G, T, A, C, C, C, T, C.

# Specialized models (BioBERT, SciBERT, Galactica)



**Figure 5:** Prompt Pre-training. Pre-training weighs all tokens equally as part of the self-supervised loss. This leads to a weak relative signal for tasks of interest, meaning model scale has to be large to work. Instruction tuning boosts performance post hoc, and can generalize to unseen tasks of interest, but it risks performance in tasks that are distant from instruction set tasks. Prompt pre-training has a weaker task of interest bias than instruction tuning but less risk of degrading overall task generality.

# Specialized models (BioBERT, SciBERT, Galactica)

- **GeLU Activation** - GeLU activations for all model sizes.
- **Context Window** - a 2048 length context window.
- **No Biases** - following PaLM, no bias in any of the dense kernels or layer norms.
- **Learned Positional Embeddings** - learned positional embeddings for the model.
- **Vocabulary** - vocabulary of 50k tokens using BPE. The vocabulary was generated from a randomly selected 2% subset of the training data.

# Specialized models (BioBERT, SciBERT, Galactica)

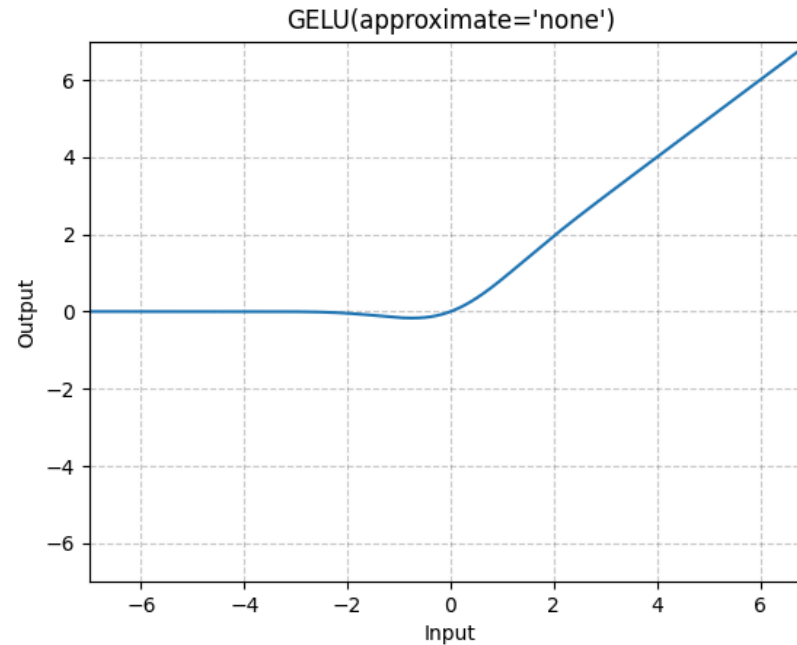
*Gaussian Error Linear Units function (GeLu)*

$$GELU(x) = x * \Phi(x)$$

Where  $\Phi(x)$  is the Gaussian function.

$$GELU(x) \approx x * \frac{1}{2} (1 + \tanh(\frac{2}{\pi} * (x + 0.044715 * x^3)))$$

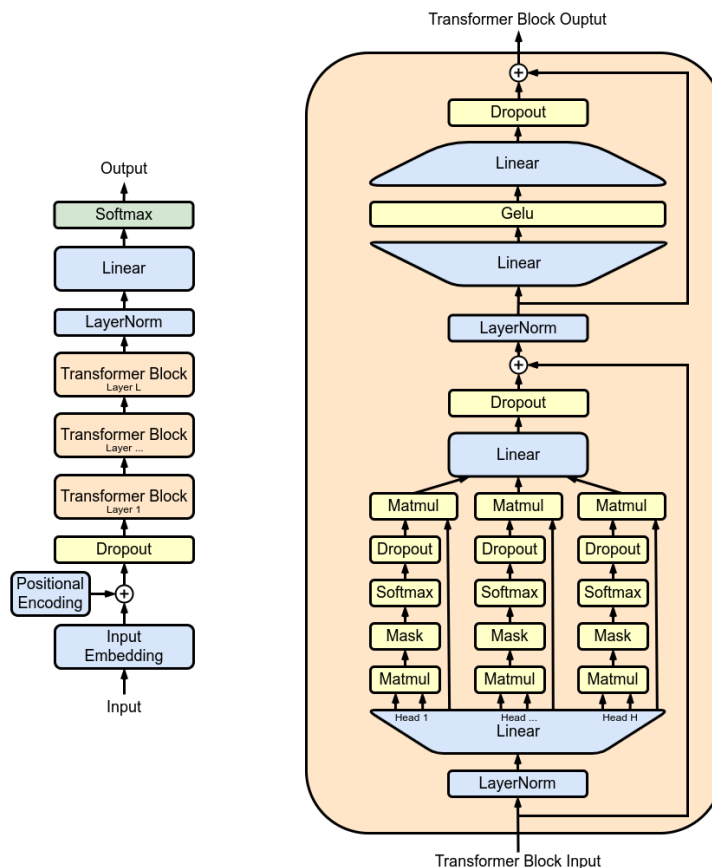
# Specialized models (BioBERT, SciBERT, Galactica)



- Allows small negative values when  $x < 0$ .
- Avoids the dying ReLU problem.

# Specialized models (BioBERT, SciBERT, Galactica)

*Why no biases?*



# Unsupervised Classification Models



# Représentations out-of-the-box: limitations

Embedding **pooling** is the process of **combining token embeddings** from an encoder model **into a single vector representing the entire input sequence**. Common methods include averaging (**mean pooling**), taking the maximum (**max pooling**), or using a **special token** like `[CLS]` or `<s>`.

# Représentations out-of-the-box: limitations



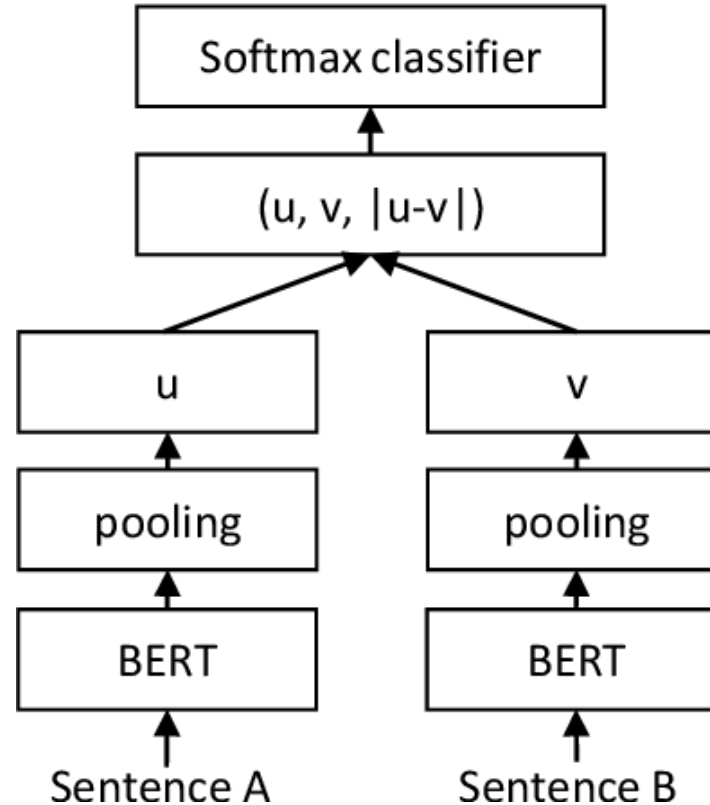
# Représentations out-of-the-box: limitations



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# Représentations out-of-the-box: limitations



**NB:** when your output pooled vectors are scaled, the cosine similarity is equal to the dot product.

# Représentations out-of-the-box: limitations

Late interaction:

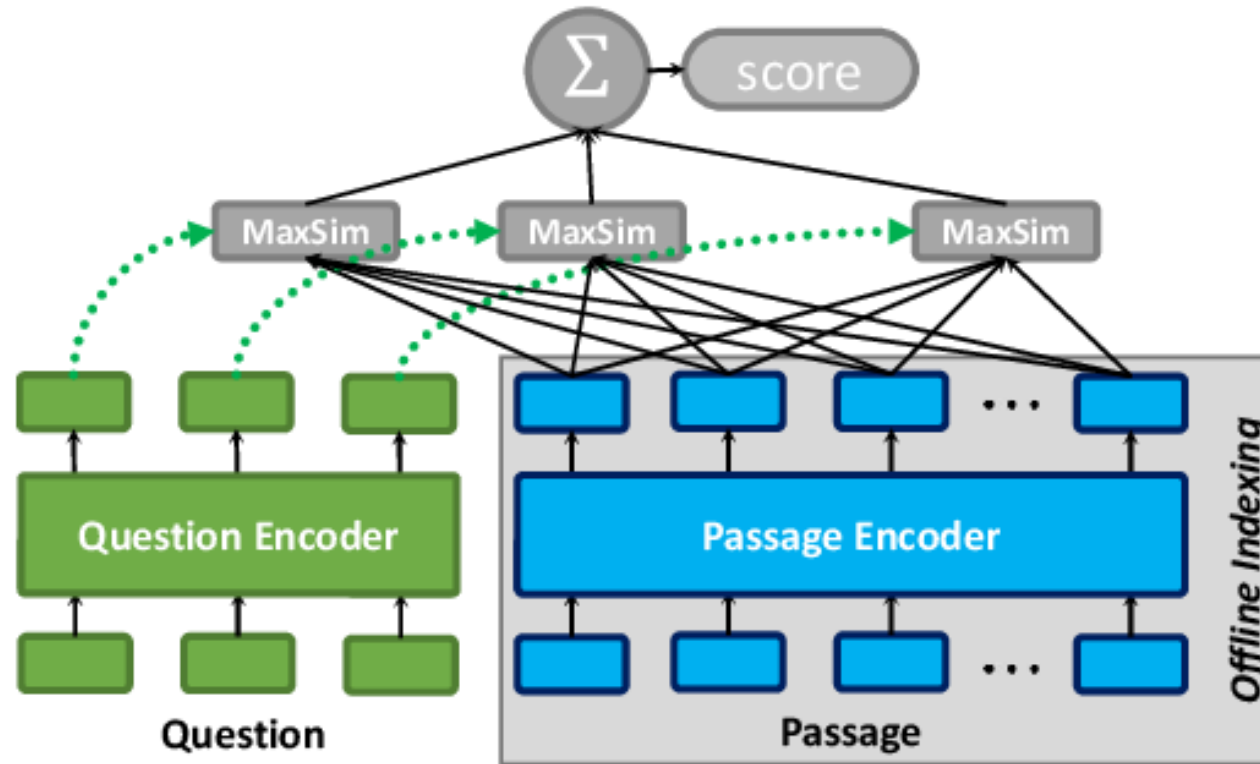


Figure 1: The late interaction architecture given

# Représentations out-of-the-box: limitations

The data is being compressed multiple times -> challenging document can be hard to embed.

Can we do better?

# SimCSE, E5, GTE...

**Contrastive learning** uses **similar data point** and **opposite ones** in order for the model build **close representations for the first ones** and **more separated ones for the latter**. [7]

- Unsupervised SimCSE: standard dropout as data augmentation
- Supervised SimCSE: use pairs in NLI datasets



# SimCSE, E5, GTE...

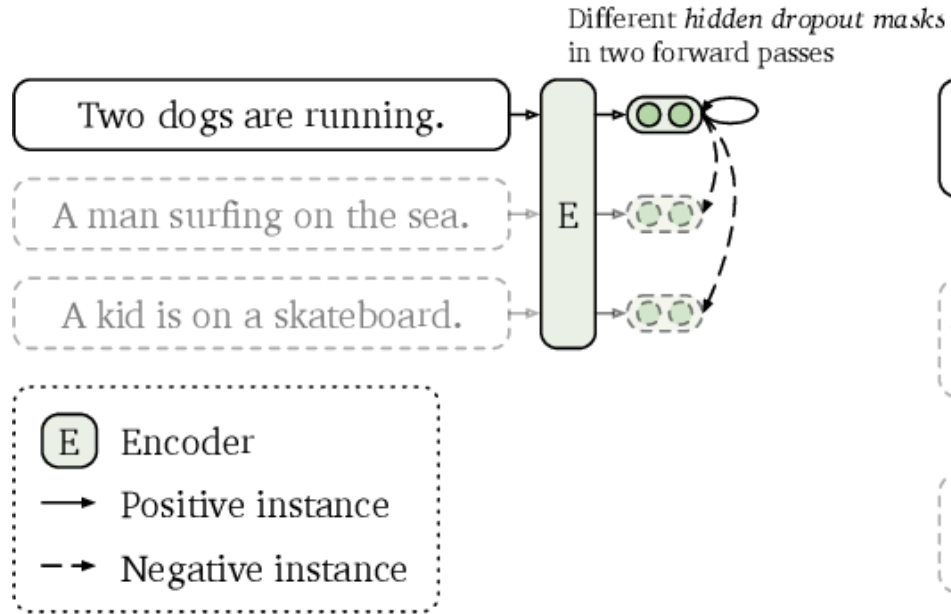
$$\mathcal{L}_{uns} = -\log \frac{\exp(\frac{\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)}{\tau})}{\sum_{j=1}^N \exp(\frac{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+)}{\tau})}$$

$$\mathcal{L}_{sup} = -\log \frac{\exp(\frac{\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)}{\tau})}{\sum_{j=1}^N \exp(\frac{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^+)}{\tau}) + \exp(\frac{\text{sim}(\mathbf{h}_i, \mathbf{h}_j^-)}{\tau})}$$

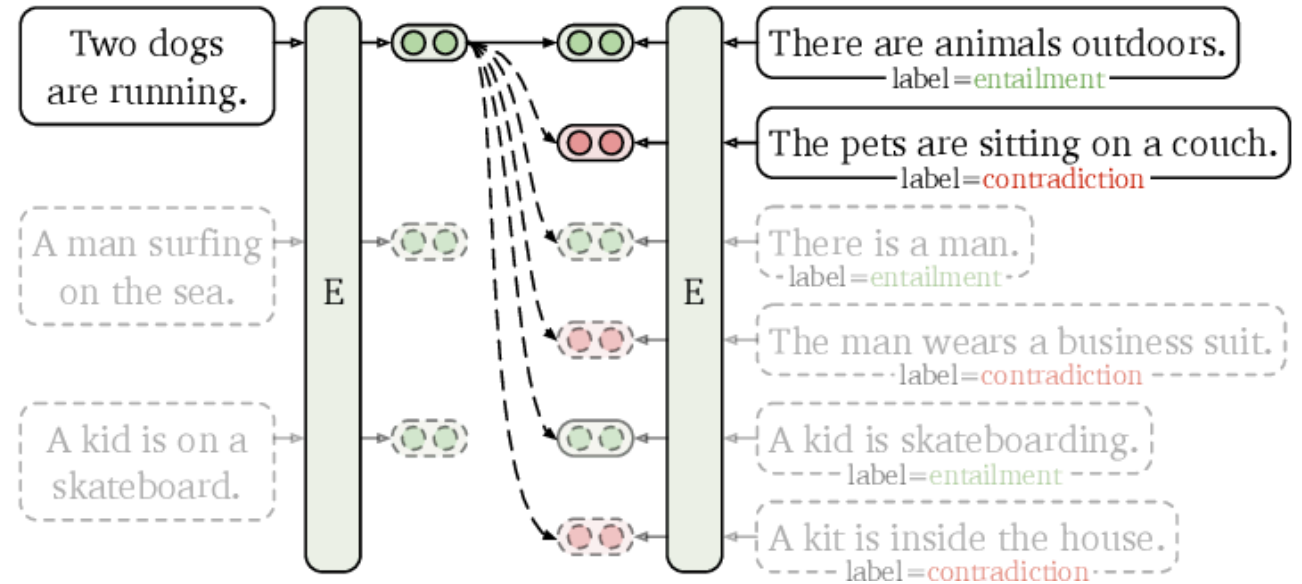
[8]

# SimCSE, E5, GTE...

(a) Unsupervised SimCSE



(b) Supervised SimCSE



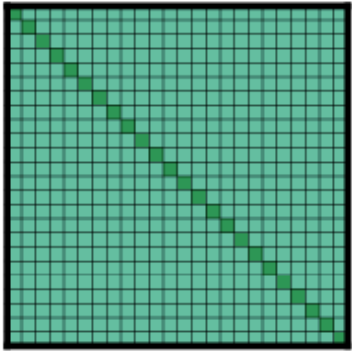
## SimCSE, E5, GTE...

**Contrastive learning mitigates anisotropy** in language models by encouraging **embeddings** to be **more uniformly distributed** in the representation space. It pulls similar embeddings closer and pushes dissimilar ones apart, preventing over-clustering and ensuring better geometric properties for downstream tasks.

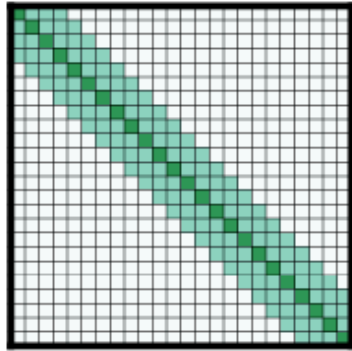
# Learning Long-Range Dependencies

# Long-range attention models

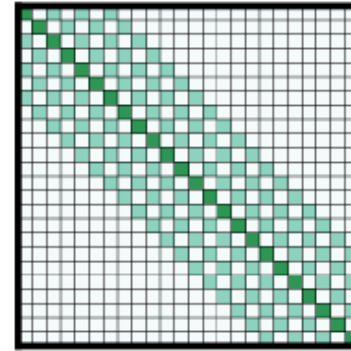
Sliding window attention: Longformer [11]



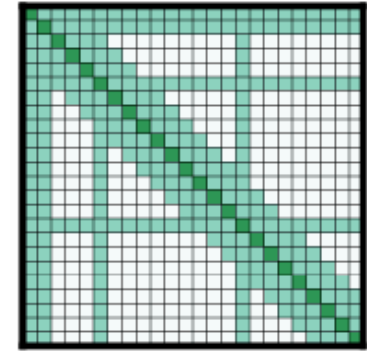
(a) Full  $n^2$  attention



(b) Sliding window attention



(c) Dilated sliding window



(d) Global+sliding window

# Long-range attention models

Sliding window attention: Mistral 7B [12]

## 2 Architectural details

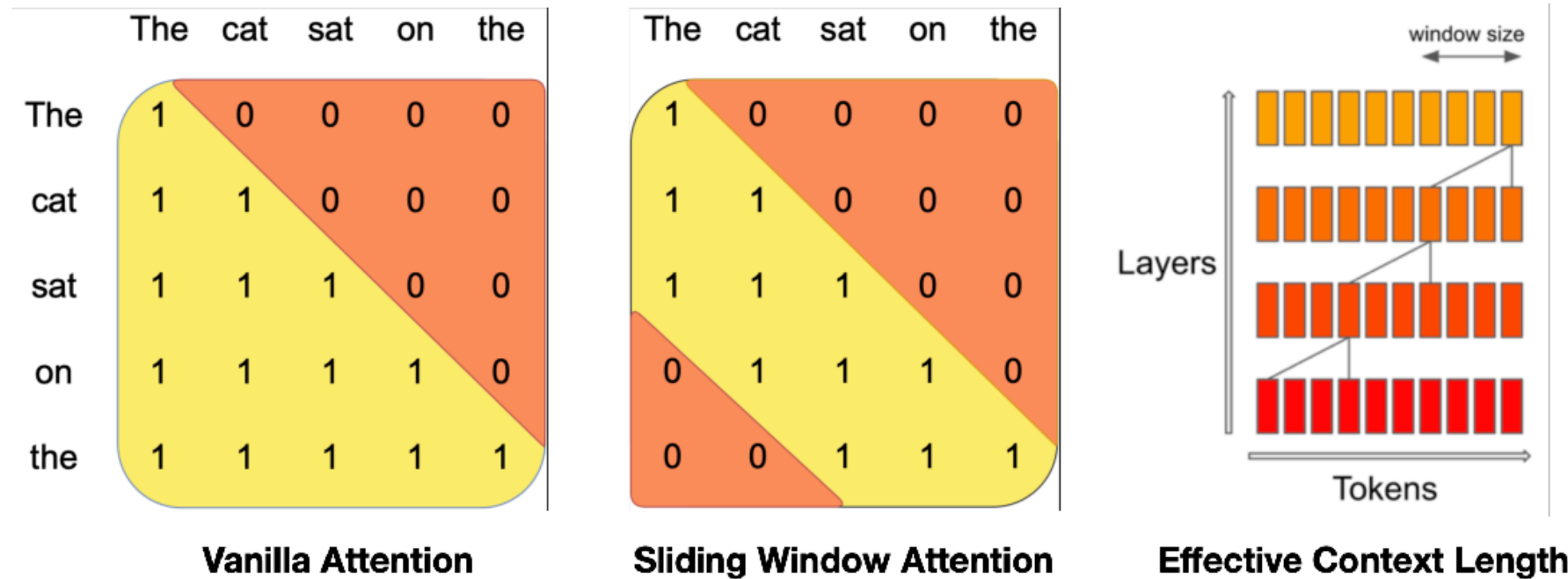
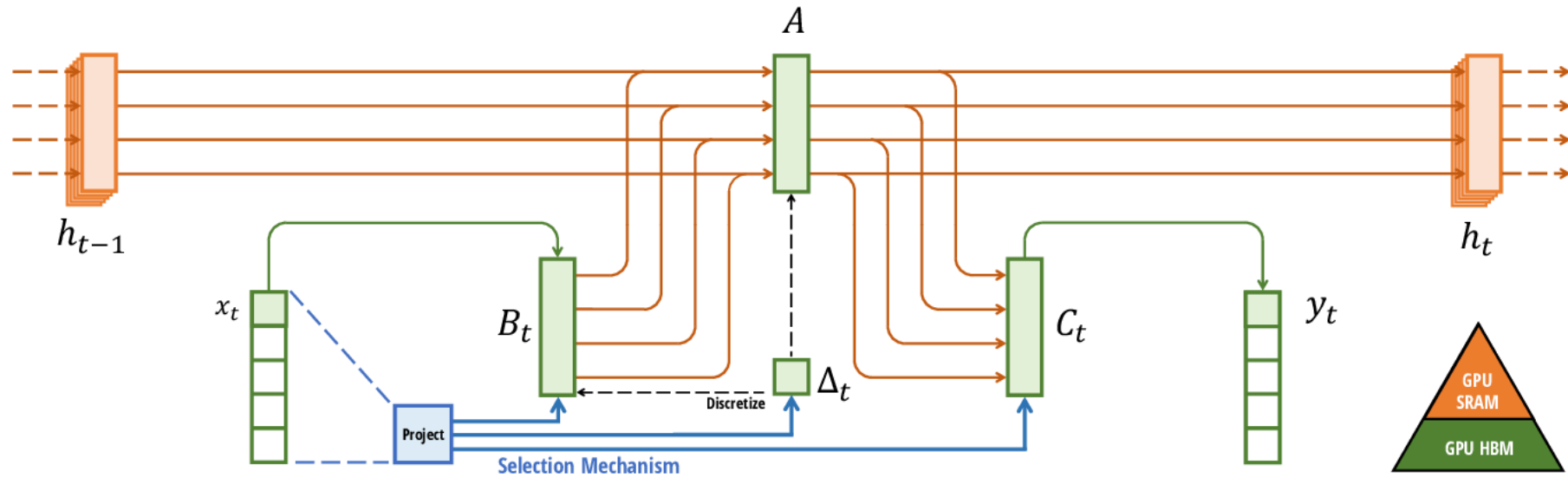


Figure 1: Sliding Window Attention. The number of operations in vanilla attention is quadratic in the sequence

# State-space models: Mamba



# Questions?



# References

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